Carl Biggers-Johanson

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LinkedIn | GitHub
Portfolio

Software Engineer | Systems · Automation · Infrastructure-Minded Development

Full-stack software engineer with a strong focus on building reliable, maintainable systems across frontend and backend applications. Experienced in TypeScript, Node.js, and API-driven architectures, with growing specialization in **system reliability, automation workflows, and scalable application design**.

I design structured systems that translate complex workflows into stable, user-facing products. My background in education and learning systems brings a unique strength in **observability, iteration, and continuous improvement**, enabling effective collaboration across teams and rapid adaptation in evolving environments.

Technical Skills

Languages & Frameworks

TypeScript, JavaScript (ES6+), Node.js, React, Express

Systems & Infrastructure (DevOps-Aligned)

CI/CD concepts, deployment workflows, environment configuration, system reliability principles, performance debugging, logging strategies

Backend & APIs

REST API design, service-layer architecture, middleware patterns, JWT authentication, API performance optimization

Frontend Engineering

Component-based architecture, state-driven UI, SPA architecture, responsive design, React Router, Vite

Databases & Data

SQL (SQLite), Prisma ORM, relational modeling, transaction handling

Tools

Git, GitHub, ESLint, VS Code

Additional Exposure

Python, Redis concepts, containerization concepts (Docker), AI-assisted development workflows

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Engineering Projects

LMS Software Demo — Full-Stack Learning Platform

GitHub: github.com/cjohanson64-netizen/lms-software-demo

- Built a full-stack application modeling real-world academic workflows (enrollment, permissions, reporting)
- Designed and implemented RESTful backend services using Node.js, Express, and Prisma
- Structured backend using service and controller layers to improve maintainability and system reliability
- Implemented role-based access control (RBAC) and secure authentication flows
- Modeled relational data to support scalable querying and consistent system behavior
- Debugged cross-system issues (frontend ↔ backend) to improve stability and reduce failure points
- Emphasized **clear system boundaries, predictable data flow, and maintainable architecture**

Predictive Pantry — Inventory & Planning App

GitHub: github.com/cjohanson64-netizen/PredictivePantry

- Developed a TypeScript + React application with structured, scalable state management
- Designed UI systems for filtering, tracking, and decision workflows
- Built reusable components supporting evolving product requirements
- Improved performance and usability through iterative debugging and refinement
- Applied system modeling techniques to reduce complexity and improve long-term maintainability
- Focused on **efficient data handling and predictable state transitions**

SightLine — Interactive Music Literacy Platform

GitHub: github.com/cjohanson64-netizen/SightLine
Live: sight-line-topaz.vercel.app

- Built an interactive web application translating complex domain data into structured UI workflows
- Designed a modular pipeline for parsing, transforming, and projecting data into frontend systems
- Developed reusable UI systems supporting real-time feedback and user interaction
- Collaborated with real users to identify issues, refine workflows, and improve reliability
- Focused on **system clarity, performance, and iterative improvement based on real-world usage**

Curriculum & Learning Systems Designer

Independent / Project-Based — Saratoga Springs, UT
2021–Present

- Designed and built structured learning systems and interactive tools for real users
- Translated complex systems into clear, repeatable workflows and user-facing processes
- Iterated on systems using feedback, usage patterns, and observed performance gaps
- Applied **continuous improvement practices**, refining systems for reliability and usability over time
- Collaborated across technical and non-technical stakeholders to align systems with real-world needs
- Leveraged AI-assisted workflows to accelerate development, testing, and documentation

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Music Educator (Choir Director)

Alpine School District — Utah
2015–Present

- Managed complex, real-time systems involving coordination, timing, and performance reliability
- Designed structured processes to ensure consistent outcomes across large groups and varying skill levels
- Continuously monitored performance, identified issues, and implemented rapid feedback loops
- Built repeatable systems that improved reliability and reduced failure points in live environments
- Developed strong communication and cross-functional collaboration skills under high-pressure conditions

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Education

Master’s Degree — Music Education
Bachelor’s Degree — Music Education